

A COMPREHENSIVE MEASURE OF WELL-BEING (Human Development Index Prediction System)

Submitted in partial fulfillment of the requirements for
Internship Project

Submitted by:
Nikhitha Devi Lakkoju

Under the guidance of:
SmartBridge Team

Table of contents

CONTENTS	PAGE NO
Abstract	3
Problem Statement	4
Features	5
Tech Stack	6
Installation Steps	7
Usage Instructions	8
ER Diagram	9-10
Screenshots	11-18
Future Scope	19

ABSTRACT

This project focuses on predicting the Human Development Index (HDI), which is a comprehensive measure of well-being. The HDI considers key factors such as life expectancy, education, and income.

In this project, machine learning techniques are used to build a predictive model using historical data. The model is trained using Linear Regression and is capable of estimating HDI values based on user inputs.

A web application is also developed using Flask, allowing users to input data and receive real-time predictions along with HDI classification categories such as Low, Medium, High, and Very High.

This project demonstrates the integration of data science and web development to solve real-world problems.

PROBLEM STATEMENT

Human Development Index (HDI) is an important indicator used to measure the overall development and well-being of a country. However, calculating HDI manually for different conditions can be time-consuming and complex.

The objective of this project is to build a system that can predict HDI based on key indicators such as life expectancy, education levels, and income using machine learning techniques.

FEATURES

- Predicts Human Development Index (HDI)
- Classifies HDI into categories (Low, Medium, High, Very High)
- User-friendly web interface
- Real-time prediction using Flask
- Data visualization using graphs

TECH STACK

Programming Language:

- Python

Libraries Used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Pickle

Web Framework:

- Flask

Tools:

- Google Colab
- VS Code

INSTALLATION STEPS

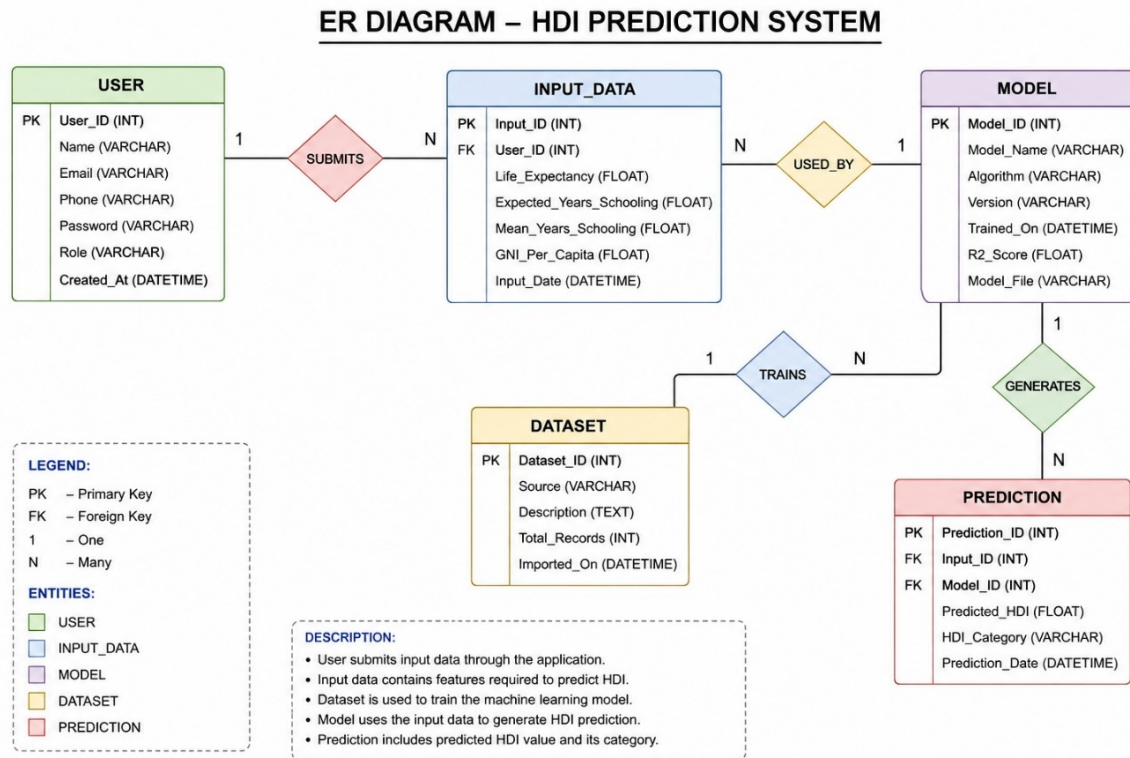
1. Install Python (3.x)
2. Install required libraries:
`pip install flask pandas numpy scikit-learn matplotlib seaborn`
3. Download the project files:
 - app.py
 - hdi_model.pkl
 - templates folder
4. Navigate to project folder:
`cd hdi_web_app`
5. Run the application:
`python app.py`

USAGE INSTRUCTIONS

1. Open browser and go to:
<http://127.0.0.1:5000/>
2. Enter the following details:
 - ☐ Life Expectancy
 - ☐ Expected Years of Schooling
 - ☐ Mean Years of Schooling
 - ☐ GNI per Capita
3. Click on "Predict"
4. The system will display:
 - ☐ Predicted HDI value
 - ☐ HDI Category

ER Diagram

Figure: Entity Relationship (ER) Diagram of HDI Prediction System



The above Entity Relationship (ER) Diagram represents the structure of the HDI Prediction System. It illustrates the relationship between different entities involved in the system.

The USER entity provides input data such as life expectancy, education details, and income through the web application. This input is stored in the INPUT_DATA entity.

The MODEL entity represents the machine learning model used for prediction, which is trained using the DATASET entity containing historical data.

The PREDICTION entity stores the output generated by the model, including the predicted HDI value and its corresponding category.

The relationships show that a user can submit multiple inputs, the model uses input data to generate predictions, and the dataset is used to train the model.

This diagram helps in understanding how data flows through the system and how different components interact with each other.

The ER diagram ensures proper data organization and helps in designing a scalable and efficient system.

SCREENSHOTS

Figure 1: HDI Distribution Histogram

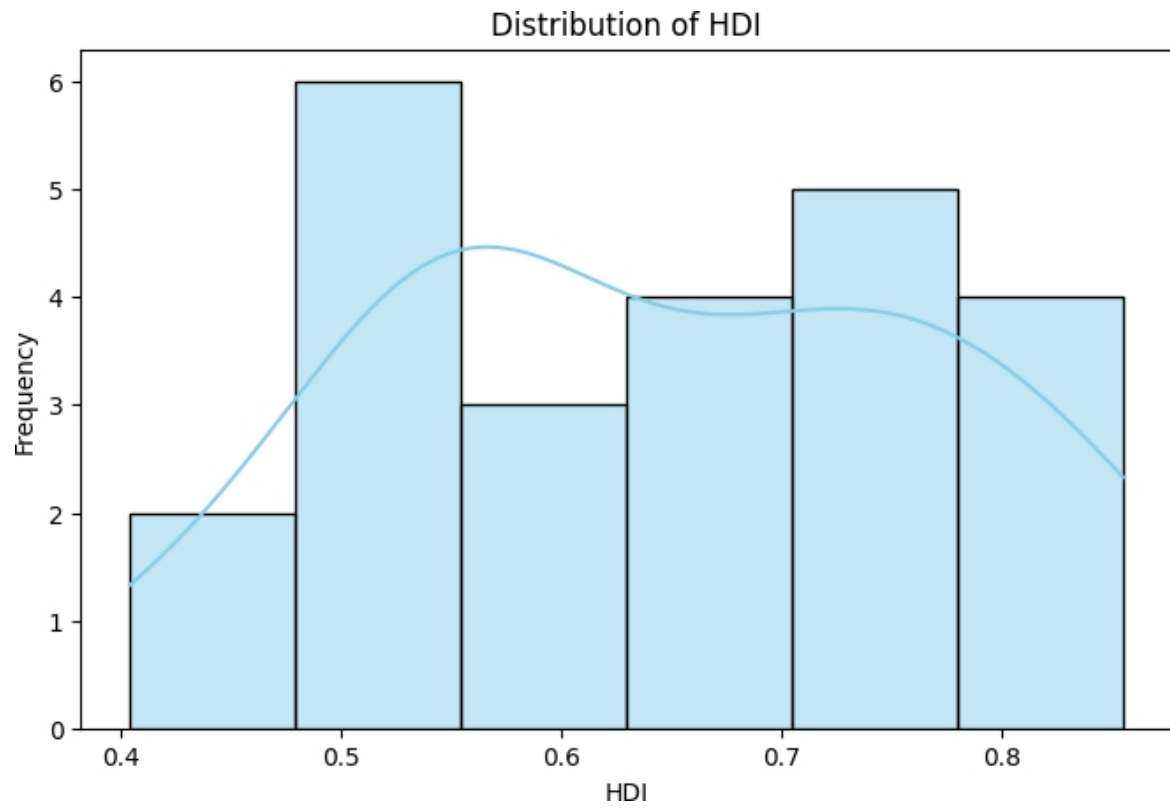


Figure 2: Correlation Heatmap

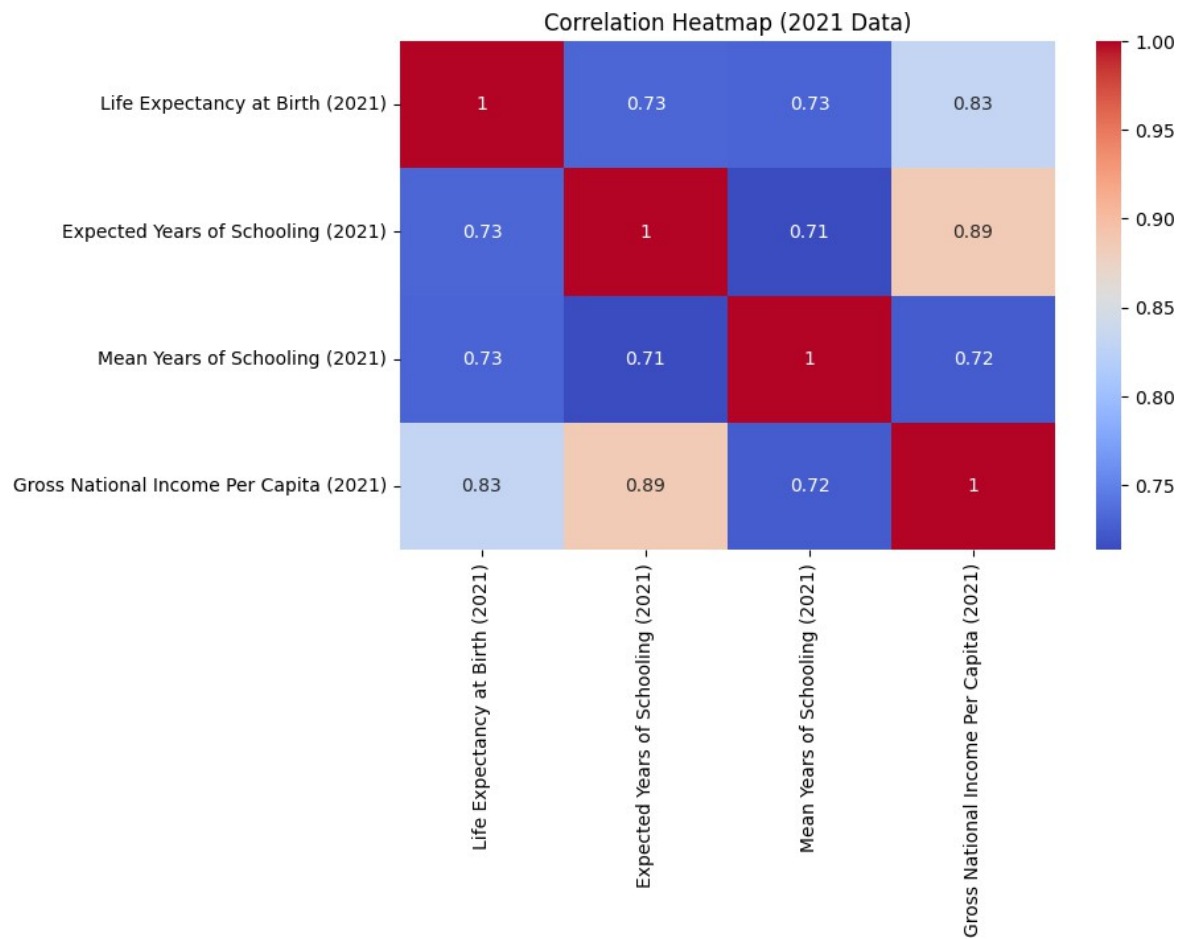


Figure 3: HDI vs Life Expectancy

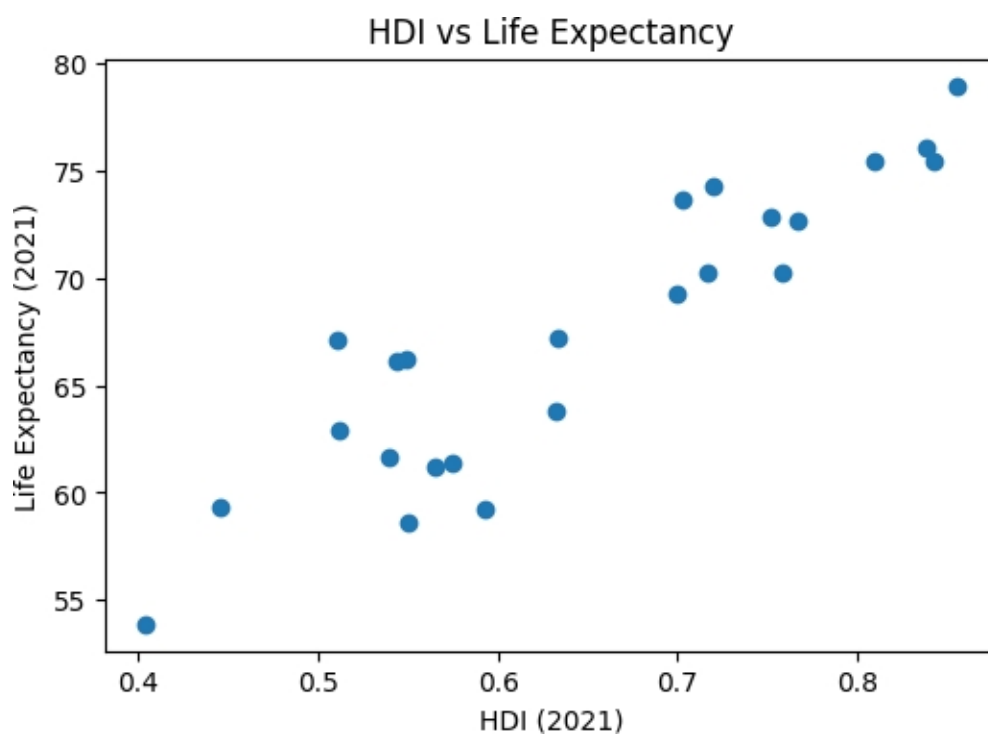


Figure 4: Web Application Interface

 **HDI PREDICTOR**

Predict Human Development Index

The Human Development Index (HDI) is a summary measure of average achievement in key dimensions of human development: a long and healthy life, being knowledgeable, and have a decent standard of living.

**Long & Healthy Life**

Estimated through Life Expectancy at Birth (minimum 20, maximum 85 years).

**Education & Knowledge**

Aggregated by combining Expected Years of Schooling and Mean Years of Schooling.

**Decent Standard of Living**

Represented by Gross National Income (GNI) per capita adjusted for PPP purchasing power.

Calculate Prediction

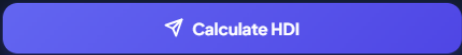
Provide country or regional statistical indicators to generate a prediction.

Life Expectancy (range: 20 – 85 years)

Expected Years of Schooling (range: 0 – 20 years)

Mean Years of Schooling (range: 0 – 15 years)

GNI per Capita (PPP \$) (range: 100 – 75,000+)



Powering Insights & Development Analytics

Calculate Prediction

Provide country or regional statistical indicators to generate a prediction.

Life Expectancy

(range: 20 – 85 years)

 75

Expected Years of Schooling

(range: 0 – 20 years)

 12

Mean Years of Schooling

(range: 0 – 15 years)

 8

GNI per Capita (PPP \$)

(range: 100 – 75,000+)

 10000

 **Calculate HDI**

Calculate Prediction

Provide country or regional statistical indicators to generate a prediction.

Life Expectancy

(range: 20 – 85 years)

 75

Expected Years of Schooling

(range: 0 – 20 years)

 12

Mean Years of Schooling

(range: 0 – 15 years)

 8

GNI per Capita (PPP \$)

(range: 100 – 75,000+)

 10000

 Calculate HDI

PREDICTED INDEX CLASS

• High Development

0.7050 / 1.0000



0.0000 (Low)

1.0000 (Very High)

 Reset Prediction

Calculate Prediction

Provide country or regional statistical indicators to generate a prediction.

Life Expectancy

(range: 20 – 85 years)

 30

Expected Years of Schooling

(range: 0 – 20 years)

 15

Mean Years of Schooling

(range: 0 – 15 years)

 10

GNI per Capita (PPP \$)

(range: 100 – 75,000+)

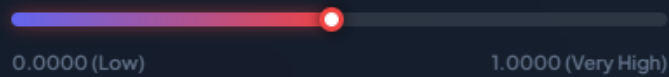
 10000

 Calculate HDI

PREDICTED INDEX CLASS

• Low Development

0.4893 / 1.0000



 Reset Prediction

Calculate Prediction

Provide country or regional statistical indicators to generate a prediction.

Life Expectancy

(range: 20 – 85 years)

 20

Expected Years of Schooling

(range: 0 – 20 years)

 10

Mean Years of Schooling

(range: 0 – 15 years)

 5

GNI per Capita (PPP \$)

(range: 100 – 75,000+)

 100000

 Calculate HDI

PREDICTED INDEX CLASS

• Medium Development

0.6406 / 1.0000

0.0000 (Low)

1.0000 (Very High)

 Reset Prediction

FUTURE SCOPE

- o Improve model accuracy using advanced algorithms
- o Add more features and real-time data
- o Deploy the application online
- o Enhance UI design
- o Add user authentication system